STUDYING RELATIONSHIP BETWEEN LOCAL GEOMETRY OF DECISION BOUNDARIES WITH NETWORK COMPLEXITY FOR ROBUSTNESS ANALYSIS

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ABSTRACT

Deep Neural networks are susceptible to adversarial attacks: if inputs are perturbed in a specific manner, it can result in misclassification. However, recent studies have shown that the robustness of the network has certain connection with the complexity of the architecture. In this paper, we investigate the distinctive influence of additional convolutional layers at the decision boundaries of several DNN architectures. To generate adversarial examples, we have utilized two different attack mechanisms, Fast Gradient Method (FGM) and One Step Spectral Attack (OSSA) using common datasets like MNIST, Fashion MNIST and Cifar 10. We have investigated the aftermaths of adding layers to the robustness of the architecture. For reasoning, we have analyzed separation width from linear class partitions and local geometry (curvature) near the decision boundary. The result shows that model complexity has significant roles in changing distances, as well as, the local features of decision boundary, which impacts the robustness of the network.